R99/B | Fixed target36 vs R89B

Native FP64 winners; same unique full18 VAL checkpoint on TRAIN/VAL/TEST.

Split	Winners	Correct	Rescued	Lost	New false	Recall (%)
TRAIN	83279	436	115	703	210	0.5235
VAL	4625	23	6	35	17	0.4973
TEST	4598	21	8	32	14	0.4567

Split	Top1 delta pp	Corrected	Harmed	Net cost delta	Regret delta pp
TRAIN	-0.349726	21879	26978	0.000939760	0.009271
VAL	-0.190123	1249	1403	0.000676602	0.004840
TEST	-0.055556	1311	1356	0.000495778	0.004847

Matched A/B: row-softmax versus fixed64 Sinkhorn, same input-only32 before four instance layers.

Original R77 CE .35 / comparison .35 / risk .02 / R-Drop .35, dropout .1; no new target/loss.

R89B main / R84A secondary historical, neither a newly matched control. Fixed R84 target36.

Not an assignment solver, VRP bound, feasibility test or B-semantics repair.

Zero-count movement alone is not success. Development-exposed TEST; no TEST/LIB selection.

R99/B | Fixed target36 vs R84A

Native FP64 winners; same unique full18 VAL checkpoint on TRAIN/VAL/TEST.

Split	Winners	Correct	Rescued	Lost	New false	Recall (%)
TRAIN	83279	436	240	101	283	0.5235
VAL	4625	23	14	2	21	0.4973
TEST	4598	21	13	5	22	0.4567

Split	Top1 delta pp	Corrected	Harmed	Net cost delta	Regret delta pp
TRAIN	0.097531	20630	19208	-0.000144223	-0.002294
VAL	-0.027160	1130	1152	0.000359409	0.002022
TEST	0.128395	1171	1067	0.000123186	-0.000354

Matched A/B: row-softmax versus fixed64 Sinkhorn, same input-only32 before four instance layers.

Original R77 CE .35 / comparison .35 / risk .02 / R-Drop .35, dropout .1; no new target/loss.

R89B main / R84A secondary historical, neither a newly matched control. Fixed R84 target36.

Not an assignment solver, VRP bound, feasibility test or B-semantics repair.

Zero-count movement alone is not success. Development-exposed TEST; no TEST/LIB selection.

R99/B | B Input Transform / Inference Timing

First128 locked TEST input rows/task; synchronized GPU wall time, three repetitions; saved evidence.

Problem	Transform ms/row	Transform + selector ms/row	Cache-hit selector ms/row	Marginal residual
TSP	1.397523	1.800659	0.397854	0.0168889
CVRP	1.792300	2.282085	0.485822	0.0165278
ATSP	0.102403	0.176832	0.074030	0.0358955
OVRP	0.121215	0.192904	0.071828	0.0210742
VRPB	0.120090	0.192935	0.071511	0.0178392
VRPL	0.120463	0.192968	0.071357	0.0117588
VRPTW	0.120821	0.192403	0.071012	0.0236063
OVRPTW	0.120503	0.193603	0.071996	0.015826
OVRPB	0.122174	0.192740	0.071989	0.0180618
OVRPL	0.120301	0.193039	0.071308	0.0144582
VRPBL	0.120240	0.191676	0.071387	0.0103952
VRPBTW	0.120442	0.191798	0.071435	0.0118685
VRPLTW	0.120597	0.191612	0.071257	0.0205909
OVRPBL	0.120477	0.193109	0.072051	0.0484095
OVRPBTW	0.120987	0.192056	0.071467	0.0210505
OVRPLTW	0.119832	0.193401	0.071603	0.0203865
VRPBLTW	0.119389	0.190899	0.070839	0.0159762
OVRPBLTW	0.120135	0.191298	0.071107	0.0157114

Feature preparation stage wall time: TRAIN/VAL 388.815s; TEST 29.825s (cache reuse allowed).

Original inputs already on GPU; online raw path has no relation_nodes/relation_node_mask.

Transform includes FP16 quantization. All18 tasks include ATSP; no solver decoder.

Excludes dataset I/O and host transfer. Cache-hit selector is not full inference or I/O-inclusive E2E.

Finite64 residuals are recorded, not proof of exact Sinkhorn convergence. Device: NVIDIA GeForce RTX 3090

R99/B | Timing recovery numerical boundary

Private validator revision; approximate messages are NEVER declared bitwise equivalent.

B: Unchanged locked weights/decision rule. Measured path=mixed, TF32 policy=default. Bitwise FP16 equality on first128 rows/task=False; original mixed-online equivalence=False. Full TEST efficacy comes from untouched cached predictions, not a demonstrated exact online run.

TEST efficacy: original locked cached predictions only. No TRAIN/TEST rerun or model selection.

LIB: unchanged original reporter, same checkpoint; its native online computation is disclosed separately.